Module 14: Macroevolution - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Under the biological species concept, two groups count as different species when they:
   - A. look slightly different from each other
   - B. simply live in different places
   - C. cannot interbreed to produce fertile offspring
   - D. have any genetic difference at all
   - Answer: C
   - Why: The biological species concept centers on reproductive isolation: no successful gene flow between the groups.
2. Two related species breed at different times of year and so never mate. This is an example of a:
   - A. prezygotic barrier
   - B. postzygotic barrier
   - C. fossil record gap
   - D. new mutation
   - Answer: A
   - Why: Prezygotic barriers act before fertilization (timing, behavior, habitat). Postzygotic barriers act after a zygote forms.
3. A mountain range splits one population into two groups that slowly diverge into separate species. This is:
   - A. sympatric speciation
   - B. no speciation at all
   - C. simple hybridization
   - D. allopatric speciation
   - Answer: D

    - Why: Allopatric speciation involves geographic separation that reduces gene flow. Sympatric speciation happens without such separation.

4. A mule, the offspring of a horse and a donkey, is healthy but sterile. This illustrates a:
    - A. prezygotic barrier
    - B. postzygotic barrier
    - C. complete lack of any isolation
    - D. brand-new species forming instantly
    - Answer: B
    - Why: Hybrid sterility is a postzygotic barrier: it acts after the zygote forms and still blocks gene flow between the parent species.